CHERN CLASSES, CHERN CHARACTER AND CHERN POLYNOMIALS OF THE DERIVED TENSOR PRODUCT OF TWO PERFECT OBJECTS WHOSE CHERN CLASSES ARE DEFINED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The author closes the c_2 computation with a sentence left to the reader; it is kept verbatim in the delivered proof (source line 9675): "which the reader can verify agrees with the formula in the statement of the lemma up to a factor of 2". This is the only step that is not written out; the binomial identity for P_p and the c_1 identity are fully proven inside the counted span.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FAG (sources/stacks-chow/pinned-original-source.tex line 9601).

Allowed standard prerequisites (editorial).

- Chern classes of a perfect object and the polynomials P_p with r(E) = P_0(E). Same chapter, chow.tex, Remark labelled remark-splitting-principle-perfect (source line 9494) and the surrounding definition of Chern classes of perfect objects. Supplies the relation between the P_p and the Chern classes c_i used at proof lines 9657-9673, and the splitting principle invoked at proof line 9655.
- The corresponding tensor product and Chern character statements for vector bundles. Same chapter, Lemma labelled lemma-chern-classes-tensor-product (source line 8567) and Lemma labelled lemma-chern-character-multiplicative (source line 8986), invoked at proof lines 9628-9630. Supply the vector bundle case which the envelope reduction would reduce the lemma to; the combinatorial P_p verification in the counted span is the author's preferred second route.
- The base situation (S, delta) for bivariant operational Chow groups. Same chapter, Situation labelled situation-setup (source line 1372), invoked in the statement. Fixes the ambient situation and the ring $A\hat{}*(X)$ in which the identities are stated; no further step of the proof uses it.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FA6: sources/stacks-chow/pinned-original-source.tex, source lines 8567–8567. Same-chapter pointer into the author environment lemma at source line 8567; invoked at proof line 9629 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9D: sources/stacks-chow/pinned-original-source.tex, source lines 8986–8986. Same-chapter pointer into the author environment lemma at source line 8986; invoked at proof line 9630 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original remark 0FAD: sources/stacks-chow/pinned-original-source.tex, source lines 9494–9494. Same-chapter pointer into the author environment remark at source line 9494; invoked at proof line 9632 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 02QL let X be locally of finite type over S. Let E and F be perfect objects of $D(\mathcal{O}_X)$ whose Chern classes are defined. Then we have*

$$c_1(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} F) = r(E)c_1(\mathcal{F}) + r(F)c_1(\mathcal{E})$$

and for $c_2(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} F)$ we have the expression

$$r(E)c_2(F)+r(F)c_2(E)+\binom{r(E)}{2}c_1(F)^2+(r(E)r(F)-1)c_1(F)c_1(E)+\binom{r(F)}{2}c_1(E)^2$$

and so on for higher Chern classes in $A^*(X)$. Similarly, we have $ch(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} F) = ch(E)ch(F)$ in $A^*(X) \otimes \mathbf{Q}$. More precisely, we have

$$P_p(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} F) = \sum\nolimits_{p_1+p_2=p} \binom{p}{p_1} P_{p_1}(E)P_{p_2}(F)$$

in $A^p(X)$.

Proof. After choosing an envelope $f : Y \to X$ such that Lf^*E and Lf^*F can be represented by locally bounded complexes of finite locally free $\mathcal{O}_X$-modules this follows by a computation from the corresponding result for vector bundles in Lemmas 0FA6 and 0F9D. A better proof is probably to use the splitting principle as in Remark 0FAD and reduce the lemma to computations in polynomial rings which we describe in the next paragraph.

Let A be a commutative ring (for us this will be the subring of the bivariant chow ring of X generated by Chern classes). Let S be a finite set together with maps $\epsilon : S \to \{\pm 1\}$ and $f : S \to A$. Define

$$P_p(S, f, \epsilon) = \sum\nolimits_{s \in S} \epsilon(s)f(s)^p$$

in A. Given a second triple (S', ϵ', f') the equality that has to be shown for P_p is the equality

$$P_p(S \times S', f + f', \epsilon\epsilon') = \sum\nolimits_{p_1+p_2=p} \binom{p}{p_1} P_{p_1}(S, f, \epsilon)P_{p_2}(S', f', \epsilon')$$

To see this is true, one reduces to the polynomial ring on variables $S \amalg S'$ and one shows that each term $f(s)^i f'(s')^j$ occurs on the left and right hand side with the same coefficient. To verify the formulas for $c_1(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} F)$ and $c_2(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} F)$ we use

the splitting principle to reduce to checking these formulae in a torsion free ring. Then we use the relationship between $P_j(E)$ and $c_i(E)$ proved in Remark 0FAD. For example

$$c_1(E \otimes F) = P_1(E \otimes F) = r(F)P_1(E) + r(E)P_1(F) = r(F)c_1(E) + r(E)c_1(F)$$

the middle equation because $r(E) = P_0(E)$ by definition. Similarly, we have

$$\begin{aligned}
&2c_2(E \otimes F) \\
&= c_1(E \otimes F)^2 - P_2(E \otimes F) \\
&= (r(F)c_1(E) + r(E)c_1(F))^2 - r(F)P_2(E) - P_1(E)P_1(F) - r(E)P_2(F) \\
&= (r(F)c_1(E) + r(E)c_1(F))^2 - r(F)(c_1(E)^2 - 2c_2(E)) - c_1(E)c_1(F) - \\
&\quad r(E)(c_1(F)^2 - 2c_2(F))
\end{aligned}$$

which the reader can verify agrees with the formula in the statement of the lemma up to a factor of 2. □